NATHAN ROBBINS

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SENIOR GIS CONSULTANT

Results-driven Senior GIS Consultant with 25 years' experience in geospatial data management, analysis, and application development for the government, environmental, and engineering industries. Adept at leveraging advanced technical solutions, scripting, and automation to optimize systems and enhance operational efficiency. Detail-oriented problem solver with a strong analytical mindset, combining research, technical support, and geodatabase management expertise. Proven leader and collaborative team player with a track record of fostering cross-functional partnerships and facilitating seamless communication across all professional levels.

KEY SKILLS

- Geographic Information Systems
- Enterprise GIS Data Services
- GIS Data Management
- GIS Data Extract, Transform, Load
- GIS Systems Integration
- GIS Solution Design
- Cloud-based GIS Workflows
- GIS Web Applications
- GIS Data Quality Assurance

Technical Proficiency: ArcGIS Dashboard, ArcGIS Enterprise, ArcGIS Experience Builder, ArcGIS Monitor, ArcGIS Online, ArcGIS Pro, ArcGIS Server, Linux Environments, Python, SAFE Software FME, SQL (Structured Query Language), AWS (Amazon Web Services), Cloud Optimized GeoTIFF

PROFESSIONAL EXPERIENCE

GEOSPATIAL DATA MANAGER, WSP, OREM, UT, 2015 - PRESENT

- Installed, configured, upgraded, and administered ArcGIS Enterprise components in Linux and Windows environments, including Portal for ArcGIS, ArcGIS Server, ArcGIS Data Store, and Web Adaptors, ensuring system reliability, performance, and adherence to GIS best practices.
- Conducted a comprehensive assessment of Newport News Shipbuilding's ArcGIS Enterprise environment, analyzing infrastructure, applications, integrations, and operational workflows, and delivered strategic recommendations and a modernization roadmap for future platform enhancements.
- Served as technical lead for the Interagency Imagery Publishing Platform (IIPP) project, overseeing the migration of enterprise thematic raster data from a standalone ArcGIS Server site to the IIPP system. Integrated remote sensing, lidar, terrain, and raster thematic data into cloud-based workflows using AWS, ensuring seamless processing and publishing of raster data.
- Utilized Python scripting and AI tools to troubleshoot and adapt scripts for batch data conversion, including transforming data from TIFF and Cloud-Optimized GeoTIFF formats to cloud raster formats, demonstrating the ability to enhance efficiency through automation.
- Published mosaic datasets as image services to ArcGIS Enterprise Portal, enhancing data accessibility and visualization for multi-agency use.
- Collaborated with DBAs and systems engineers to resolve complex issues, such as troubleshooting SDE errors in the AWS Kubernetes environment caused by system updates. Coordinated discussions with ESRI to identify root causes and prevent future disruptions, demonstrating strong problem-solving and teamwork skills.

- Designed and implemented ESRI Experience Builder solutions, including the Forest Service Interactive Visitor Map (IVM), replacing custom-developed applications with commercial off-the-shelf (COTS) solutions and saving the agency hundreds of thousands of dollars annually.
- Streamlined workflows for the IVM project, reducing FME processes from 37 to 10 and map services from 10 to 5, cutting server load in half and significantly improving efficiency while reducing data duplication and redundancy.
- Developed and refined Arcade expressions for AGOL popups as part of the IVM project, creating feature sets to display only relevant data fields with values, improving usability and data presentation.

GIS MANAGER, AECOM, RALEIGH, NC, 2007 - 2015
- Delivered technical solutions for high-priority projects, including pipeline and emergency response efforts, leveraging GIS expertise to save thousands in revenue and improve operational efficiency.
- Automated repetitive tasks using Python scripts, enhancing productivity and reducing manual effort.
- Designed and implemented mobile data collection systems for emergency response, improving data accuracy and accessibility.
- Collaborated with engineering firms and third-party contractors to facilitate seamless GIS and AutoCAD data conversion.

GIS PROJECT MANAGER, GEOGRAPHIC TECHNOLOGIES GROUP, GOLDSBORO, NC, 2005 - 2007
- Managed end-to-end GIS projects for local government agencies, including database design, data validation, and strategic planning.
- Executed large-scale AutoCAD-to-GIS conversion projects, providing training and technical support to ensure successful implementation.
- Deployed ArcSDE and SQL Server database solutions, delivering training and ongoing support for database management.

GIS TECHNICIAN, EARTH TECH, SHAW AIR FORCE BASE, SUMTER, SC, 2004 - 2005
- Developed a standardized GPS data dictionary adopted across the U.S. Air Force, ensuring consistency in data collection.
- Created and maintained geographic databases to support mission-critical operations.
- Provided onsite technical support and troubleshooting for GIS software and applications.

GIS TECHNICIAN, CITY OF CHESAPEAKE, VA, 2002 - 2004
- Developed maps, graphics, and reports to support city managers and inform the public.
- Assisted in natural disaster response by creating emergency plans for recovery and control.
- Improved operational efficiency by documenting and standardizing internal processes.

EDUCATION

BS, Geographic Information Systems, GPA: 3.7, Brigham Young University, Provo, UT, 2002
Geographic Information Systems, Graduate Courses (1 Yr.), Penn State University, College Station, PA
Technology Systems, Digital Communications, Graduate Courses (1 Yr.), East Carolina University, Greenville, NC

CERTIFICATIONS

Geographic Information Systems Professional (GISP) – GIS Certification Institute, Certification #162811

PROFESSIONAL/ COMMUNITY CONTRIBUTIONS

- **Recognized with Chief's Honor Award** from the Forest Service in 2023 for delivering the Non-Wildfire Crisis Strategy BIL and IRA programs with ingenuity.
- **Recognized with "Making a Difference" Award** for contribution to resolving Michigan oil spill and implementing a mobile data collection platform for environmental soil boring data.
- **Fulfilled various program leadership and support roles** such as group president and BSA Scoutmaster, as an actively committed member of community church.
- **Served 2-year proselyting** / service mission in Argentina.